

# RBE595: Hands-On Autonomous Aerial Robotics

## P1b: Non-stinky Unscented Kalman Filter for Attitude Estimation

Shreyas Devdatta Khobragade  
Department of Robotics Engineering  
Worcester Polytechnic Institute  
Email: skhobragade@wpi.edu

Brijan Vaghasiya  
Department of Robotics Engineering  
Worcester Polytechnic Institute  
Email: bcvaghasiya@wpi.edu

**Abstract**—This paper investigates attitude estimation from a 6-DoF Inertial Measurement Unit (IMU) using several complementary approaches and the Unscented Kalman Filter (UKF). Baseline methods include gyroscope integration, accelerometer inclination, and a simple complementary filter, along with the Madgwick algorithm, a gradient-descent-based orientation estimator. Ground truth orientation is provided by a Vicon motion-capture system with temporal alignment via spherical linear interpolation (Slerp). While accelerometer-only estimates are noisy and pure integration suffers from drift, the complementary and Madgwick filters improve stability, with the latter showing robustness under dynamic motion. Building on these methods, the UKF framework is presented as a principled alternative to the Extended Kalman Filter (EKF), avoiding Jacobian linearization by propagating sigma points through nonlinear quaternion dynamics. The combined evaluation highlights the trade-offs between lightweight filtering and the more rigorous UKF approach for precise and reliable orientation estimation in aerial and robotic systems.

**Index Terms**—IMU, attitude estimation, gyroscope integration, accelerometer inclination, complementary filter, Madgwick filter, Unscented Kalman Filter, Vicon, Slerp.

### I. INTRODUCTION

Accurate estimation of 3-D orientation is fundamental in aerospace, robotics, navigation, and human motion capture, where system stability and control rely on precise attitude knowledge. Low-cost IMUs provide three-axis gyroscope and accelerometer measurements, but each modality has complementary weaknesses: gyroscopes measure angular velocity well over short durations yet suffer integration drift from bias and noise, while accelerometers observe gravity for roll and pitch estimation but are corrupted by translational accelerations and cannot measure yaw. Robust orientation estimation therefore requires sensor fusion. This work addresses the Phase 1 problem statement P0 using ArduIMU+ V2 data with four methods: (a) gyroscope integration with Vicon-based initialization, (b) accelerometer-only orientation under a pure rotation assumption, (c) a complementary filter combining gyro and accelerometer estimates, and (d) the Madgwick filter. The Madgwick algorithm applies quaternion-based gradient descent to align estimated and measured gravity, rejecting

disturbances from linear acceleration and correcting gyro drift without the high-frequency tuning needed in complementary filters. Compared to Euler-angle representations, quaternions eliminate gimbal lock, enable smooth interpolation (e.g., SLERP), and improve numerical stability for real-time use. While complementary and gradient-descent filters reduce drift and noise, they remain heuristic approaches. Classical Kalman filtering provides an optimal solution for linear Gaussian systems, but nonlinear quaternion dynamics challenge its assumptions. The Extended Kalman Filter (EKF) mitigates this through Jacobian linearization, introducing approximation errors and computational cost. The Unscented Kalman Filter (UKF) avoids these limitations by propagating sigma points through the nonlinear dynamics, preserving higher-order accuracy in mean and covariance. This makes the UKF a principled framework for accurate, robust, and real-time quaternion-based attitude estimation. Ground-truth orientation from Vicon motion capture is used for synchronization, spherical linear interpolation, and evaluation of all methods. This report details the mathematical formulations, implementation, and comparative performance analysis of the four baseline filters alongside the UKF framework.

### II. DATA AND TIMESTAMP SYNCHRONIZATION

#### A. Dataset

We use the Phase1 dataset composed of synchronized pairs:

- Phase1/Data/Train/IMU/imuRawN.mat containing vals (columns represent  $[a_x, a_y, a_z, \omega_z, \omega_x, \omega_y]^T$ ) and ts timestamps (1×M).
- Phase1/Data/Train/Vicon/viconRotN.mat containing rots (3×3×K rotation matrices in Z-Y-X order) and ts timestamps (1×K).

Instrument calibration parameters are read from Phase1/IMUParams.mat.

#### B. Sensor calibration and unit conversion

The raw measurements are transformed into physical units. Accelerometer units (m/s<sup>2</sup>) are computed per axis as:

$$\tilde{a}_x = \frac{a_x + b_{a,x}}{s_x},$$

(and similarly for  $a_y, a_z$ ). Gyroscope raw counts are converted to rad/s:

$$\tilde{\omega} = \left( \frac{3300}{1230} \right) \cdot \frac{\pi}{180} \cdot 0.3 \cdot (\omega - b_g),$$

where  $b_g$  is estimated from the first 200 samples.

### C. Time Synchronization and Interpolation

The IMU and Vicon data are aligned over the common window

$$t_{\min} = \max(\min t^{\text{imu}}, \min t^{\text{vic}}), \quad (1)$$

$$t_{\max} = \min(\max t^{\text{imu}}, \max t^{\text{vic}}). \quad (2)$$

A match is accepted if

$$|t^{\text{imu}} - t^{\text{vic}}| \leq \varepsilon, \quad \varepsilon = 0.10 \text{ s}.$$

The Vicon rotation matrices were interpolated to match the IMU timestamps using **Spherical Linear Interpolation (SLERP)** on unit quaternions. Given two quaternions  $q_1, q_2$  and interpolation factor  $\alpha \in [0, 1]$ , SLERP is defined as:

$$\text{SLERP}(q_1, q_2; \alpha) = \frac{\sin((1 - \alpha)\theta)}{\sin \theta} q_1 + \frac{\sin(\alpha\theta)}{\sin \theta} q_2,$$

where  $\theta = \arccos(q_1 \cdot q_2)$ .

In practice, this interpolation was performed using the built-in `scipy.spatial.transform.Slerp` function in Python, which allows Vicon rotation matrices to be aligned with IMU timestamps.

## III. METHODOLOGY

### A. Gyroscope integration

The gyroscope provides angular velocity measurements in the body frame. To obtain Euler angles, these body rates must first be converted to Euler angle rates. At each time step, the conversion is performed using the mapping:

$$\dot{e} = J(\phi, \theta) \omega, \quad J(\phi, \theta) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix}.$$

With timestep  $\Delta t$ , integrate:

$$e_{k+1} = e_k + J(e_k) \omega_k \Delta t.$$

where  $\omega = [\omega_x, \omega_y, \omega_z]^T$  represents the angular velocity in the body frame, and  $\phi, \theta, \psi$  are the Euler angle rates. In simpler terms, after the Euler rates are obtained in world frame, the orientation can be updated at the next time step using discrete numerical integration:

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_{t+\Delta t} = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_t + \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_t \cdot \Delta t. \quad (3)$$

For the initial step, the starting orientation is initialized using the mean of a few hundred Vicon ground truth measurements:

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_{t_0} = \frac{1}{N} \sum_{k=1}^N x_{\text{vicon},k}, \quad (4)$$

where  $N$  represents the number of initial readings chosen. For instance, in dataset 1, the drone was assumed to be at rest up to approximately the 840th reading. Similarly, for other datasets, one can examine the initial period when the system is stationary and use the corresponding measurements to initialize the orientation.

### B. Accelerometer inclination

Accelerometer-based attitude is calculated by projecting the acceleration vector onto the IMU body axes, using gravity (pointing downward along  $-Z$ ) as reference. Roll and pitch from accelerometer:

$$\phi_a = \text{atan2}(a_y, \sqrt{a_x^2 + a_z^2}), \quad \theta_a = \text{atan2}(a_x, \sqrt{a_y^2 + a_z^2}).$$

Yaw estimation is limited due to the Z-axis symmetry of gravity, but a small tilt of the IMU enables an approximate calculation. Accelerometer-based attitude is obtained by projecting the acceleration vector onto the IMU body axes with gravity ( $-Z$ ) as reference. Yaw ( $\psi$ ) can be computed using different methods for this assignment, we use:

$$\psi = \arctan \frac{\sqrt{a_x^2 + a_y^2}}{a_z}$$

This works well for our purposes despite other possible formulas.

### C. Low-pass filtering

To reduce high-frequency noise and rapid disturbances in the IMU measurements, an exponential low-pass filter is applied:

$$x_f[k] = (1 - \gamma)x_f[k-1] + \gamma x[k],$$

where  $x[k]$  is the raw sensor measurement,  $x_f[k]$  is the filtered output, and  $\gamma \in (0, 1]$  controls the filter strength. Accelerometers perform best at low-speed or quasi-static conditions, as dynamic motions introduce errors due to translational accelerations. The low-pass filter helps emphasize the slower, gravity-dominated components of the acceleration signal, improving the reliability of roll and pitch estimation.

### D. High-pass Filtering

To remove low-frequency drift and bias from gyroscope measurements, a high-pass filter is applied. This emphasizes fast-changing components of the signal while suppressing slow drift:

$$y_f[k] = \alpha(y_f[k-1] + y[k] - y[k-1]),$$

where  $y[k]$  is the raw gyro measurement,  $y_f[k]$  is the filtered output, and  $\alpha \in (0, 1]$  controls the filter strength. Gyroscopes are reliable at capturing high-speed angular motion but suffer from long-term drift. The high-pass filter allows the gyro to retain responsiveness to fast rotations while minimizing slow drift effects.

### E. Complementary filter

The complementary filter combines the strengths of accelerometer and gyroscope measurements to produce a more robust orientation estimate. Accelerometers are reliable at low-speed or static conditions, while gyroscopes perform well at high-speed motion but suffer from drift. The complementary filter fuses both sources by applying a low-pass filter to the accelerometer and a high-pass filter to the gyroscope:

$$\hat{\theta}[k] = \alpha \theta_{\text{acc}}[k] + (1 - \alpha) (\hat{\theta}[k-1] + \omega[k] \Delta t),$$

where  $\hat{\theta}[k]$  is the estimated angle,  $\theta_{\text{acc}}[k]$  is the accelerometer-based angle,  $\omega[k]$  is the gyro rate,  $\Delta t$  is the time step, and  $\alpha \in (0, 1)$  controls the balance between sensors. For this assignment,  $\alpha$  was selected as 0.8 to balance accelerometer and gyroscope contributions, and the low-pass filter coefficient  $\gamma$  for the accelerometer was set to 0.1. This fusion process is illustrated in Figure 1.

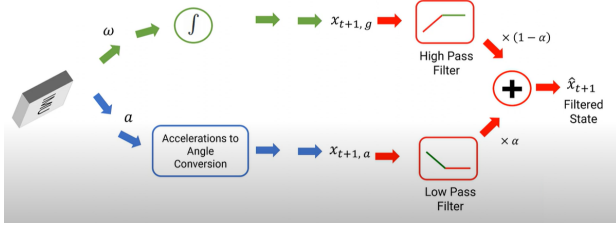


Fig. 1: Complementary filter combining accelerometer and gyroscope measurements.

### F. Madgwick Filter

The **Madgwick filter** is a computationally efficient and robust orientation estimation algorithm that fuses gyroscope and accelerometer data (optionally magnetometer) using a *quaternion-based representation*. Unlike Euler angles, this approach avoids singularities such as gimbal lock and maintains numerically stable real-time orientation tracking. The filter integrates gyroscopic measurements to propagate orientation, while applying a corrective feedback derived from accelerometer data using a gradient descent optimization that minimizes the error between expected and measured gravity vectors.

1) *Quaternion Derivative from Gyroscope*: The gyroscope provides angular velocity measurements, which are used to compute the derivative of the orientation quaternion as:

$$\dot{q}_\omega = \frac{1}{2} q \otimes \omega \quad (5)$$

where:

- $q$  is the current orientation quaternion,
- $\omega = [0, \omega_x, \omega_y, \omega_z]$  is the angular velocity expressed as a pure quaternion,
- $\otimes$  denotes quaternion multiplication.

This equation yields the rate of change of orientation due solely to gyroscopic data.

2) *Jacobian Matrix  $J$* : In the gradient descent optimization process, the Jacobian matrix  $J$  quantifies how small changes in the quaternion  $q$ , expected gravity vector  $\hat{g}$ , and normalized accelerometer measurement  $\hat{a}$  affect the objective function  $f$ . This matrix is essential for computing the gradient used in the correction step. The gradient of the objective function is computed as:

$$\nabla f(\hat{q}_{\text{est},t}, \hat{g}, \hat{a}_{t+1}) = J^T(\hat{q}_{\text{est},t}, \hat{g}, \hat{a}_{t+1}) f(\hat{q}_{\text{est},t}, \hat{g}, \hat{a}_{t+1}) \quad (6)$$

where:

- $f(q, \hat{g}, \hat{a})$  measures the difference between the expected gravity vector and accelerometer measurement,
- $J$  is the Jacobian matrix representing sensitivities of the error function to changes in  $q$ ,  $\hat{g}$ , and  $\hat{a}$ .

3) *Accelerometer-Based Correction via Gradient Descent*: To mitigate gyroscopic drift, a correction term is applied based on the normalized gradient of the objective function:

$$\text{Correction} = -\beta \cdot \frac{\nabla f}{\|\nabla f\|} \quad (7)$$

where:

- $\beta$  is a filter gain controlling the correction strength,
- $\nabla f$  is the gradient obtained from the Jacobian matrix as above.

4) *Final Quaternion Update*: The estimated quaternion derivative is the sum of the gyroscope-based propagation and the accelerometer-based correction:

$$\dot{q}_{\text{est}} = \dot{q}_\omega + \text{Correction} = \dot{q}_\omega - \beta \cdot \frac{\nabla f}{\|\nabla f\|} \quad (8)$$

The quaternion is then updated by integrating this rate over the time step  $\Delta t$ :

$$q_{t+1} = q_t + \dot{q}_{\text{est}} \cdot \Delta t \quad (9)$$

Finally, the quaternion is normalized to ensure it remains a valid rotation:

$$q_{t+1} \leftarrow \frac{q_{t+1}}{\|q_{t+1}\|} \quad (10)$$

5) *Gain Tuning*: The filter gain  $\beta$  was empirically set to 0.22 to balance responsiveness and noise robustness. Lower values of  $\beta$  provide smoother tracking but slower drift correction, whereas higher values enable faster convergence with increased noise sensitivity.

### G. Unscented Kalman Filter

1) *State Representation*: The system state is defined as

$$x = [q_w \ q_x \ q_y \ q_z \ \omega_x \ \omega_y \ \omega_z]^T \quad (11)$$

where  $q = [q_w, q_x, q_y, q_z]^T$  is the unit quaternion describing orientation, and  $\omega = [\omega_x, \omega_y, \omega_z]^T$  is the angular velocity vector. The state covariance  $P$ , process noise covariance  $Q$ , and measurement noise covariance  $R$  parameterize the uncertainty in dynamics and sensor measurements.

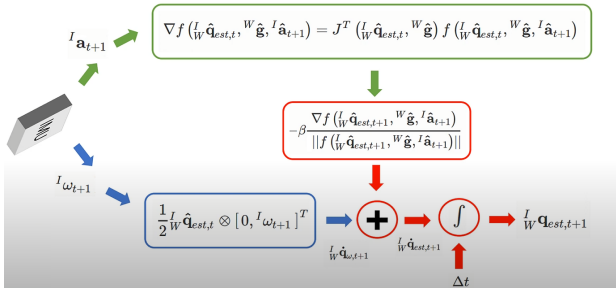


Fig. 2: Madgwick filter combining accelerometer and gyroscope measurements.

2) *Sigma Point Generation*: Unlike the EKF, the UKF does not linearize system equations. Instead, it generates a set of sigma points that capture the mean and covariance of the state distribution. The covariance matrix is decomposed via Cholesky factorization:

$$P_{k-1} = S^T S \quad (12)$$

The sigma point disturbances are defined as

$$W_{i,i+n} = \pm \sqrt{n \cdot (P_{k-1} + Q)} \quad (13)$$

Each disturbance is mapped to a quaternion and angular velocity perturbation:

$$\chi_i = \begin{bmatrix} q_{k-1} \otimes q_W \\ \omega_{k-1} + \omega_W \end{bmatrix} \quad (14)$$

where  $q_W$  is obtained from the rotation vector part of  $W_i$ .

3) *Propagation Through Process Model*: Each sigma point is propagated through the nonlinear process model:

$$Y_i = f(\chi_i) \quad (15)$$

For rotational dynamics, this includes quaternion integration:

$$q_\Delta = [\cos(\frac{\alpha_\Delta}{2}), \mathbf{e}_\Delta \sin(\frac{\alpha_\Delta}{2})] \quad (16)$$

with  $\alpha_\Delta = \|\omega\| \Delta t$  and  $\mathbf{e}_\Delta = \omega / \|\omega\|$ . Thus,

$$Y_i = \begin{bmatrix} q_{k-1} \otimes q_W \otimes q_\Delta \\ \omega_{k-1} + \omega_W \end{bmatrix} \quad (17)$$

4) *Mean and Covariance Computation*:

a) *Quaternion Mean*: Since quaternions lie on the unit sphere, a direct arithmetic mean is invalid. Instead, the mean quaternion is computed via intrinsic gradient descent. Starting with an initial guess  $q^{(0)}$ , the algorithm iteratively reduces the average rotation error until convergence:

$$q^{(t+1)} = \exp(\bar{e}) \otimes q^{(t)} \quad (18)$$

where  $\bar{e}$  is the mean rotation vector error across all sigma points.

b) *Covariance*: The covariance is estimated from sigma point deviations:

$$P_k^- = \frac{1}{2n} \sum_{i=1}^{2n} W_i' W_i'^T \quad (19)$$

with

$$W_i' = \begin{bmatrix} \log(q_i \otimes \bar{q}^{-1}) \\ \omega_i - \bar{\omega} \end{bmatrix} \quad (20)$$

5) *Measurement Update*: Each predicted sigma point is mapped into measurement space:

$$Z_i = h(Y_i) \quad (21)$$

The mean predicted measurement is:

$$\bar{z} = \frac{1}{2n} \sum_{i=1}^{2n} Z_i \quad (22)$$

The innovation covariance is:

$$P_{\nu\nu} = P_{zz} + R \quad (23)$$

The cross-covariance between state and measurement is:

$$P_{xz} = \frac{1}{2n} \sum_{i=1}^{2n} W_i' (Z_i - \bar{z})^T \quad (24)$$

The Kalman gain is:

$$K_k = P_{xz} P_{\nu\nu}^{-1} \quad (25)$$

Finally, the state and covariance are updated as:

$$\hat{x}_k = \hat{x}_k^- + K_k \nu_k \quad (26)$$

$$P_k = P_k^- - K_k P_{\nu\nu} K_k^T \quad (27)$$

where  $\nu_k = z_k - \bar{z}$  is the innovation.

6) *Measurement Model and Noise Covariances*: The covariance matrices were initialized as diagonal matrices:

$$P = \text{diag}(2 \times 10^{-2}, 2 \times 10^{-2}, 2 \times 10^{-2}, 2 \times 10^{-3}, 2 \times 10^{-3}, 2 \times 10^{-3}), \quad (28)$$

$$Q = \text{diag}(10^{-3}, 10^{-3}, 10^{-3}, 10^{-6}, 10^{-6}, 10^{-6}), \quad (29)$$

$$R = \text{diag}(0.68, 0.68, 0.68, 0.057, 0.057, 0.057). \quad (30)$$

Here,  $P$  is the initial state covariance,  $Q$  is the process noise covariance, and  $R$  is the measurement noise covariance.

#### H. Summary of UKF Steps

- 1) Compute sigma points from  $P_{k-1} + Q$ .
- 2) Map sigma points to valid quaternions and angular velocities.
- 3) Propagate through process model.
- 4) Compute mean quaternion with intrinsic gradient descent.
- 5) Compute predicted covariance.
- 6) Transform sigma points into measurement space.
- 7) Compute innovation covariance and cross-covariance.
- 8) Calculate Kalman gain.
- 9) Update state and covariance.

## IV. RESULTS

All the graphs plotted from Datasets 1 to 6 are provided below. Each dataset includes roll, pitch, and yaw angles derived from the following sources: Vicon measurements, gyroscope, accelerometer, complementary filter, madgwick filter and the unscented kalman filter results. The video outputs from rotplot.py are included in this link.

### A. Train Dataset

Figures 3 to 8

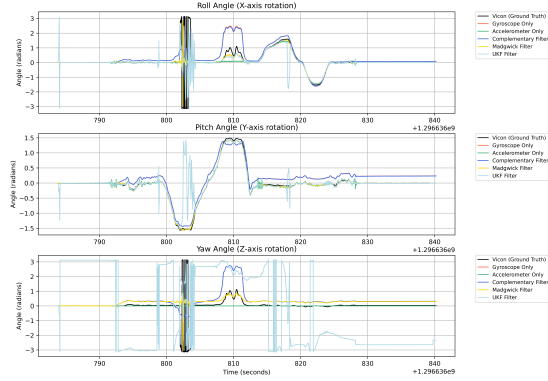


Fig. 3: Orientation comparison for Dataset 1: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

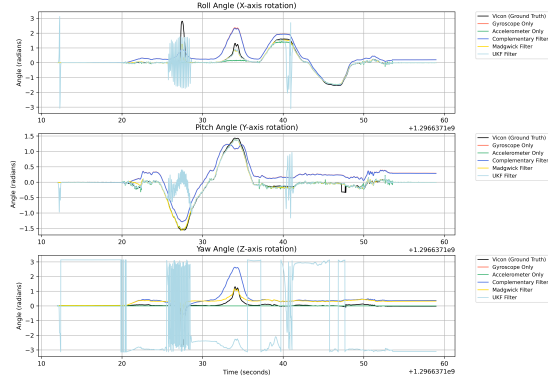


Fig. 4: Orientation comparison for Dataset 2: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

### B. Test Dataset

Figures 9 to 12

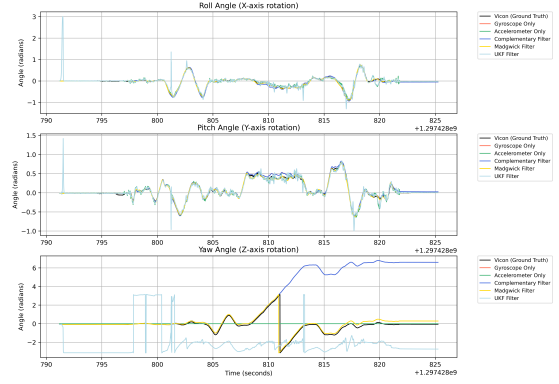


Fig. 5: Orientation comparison for Dataset 3: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

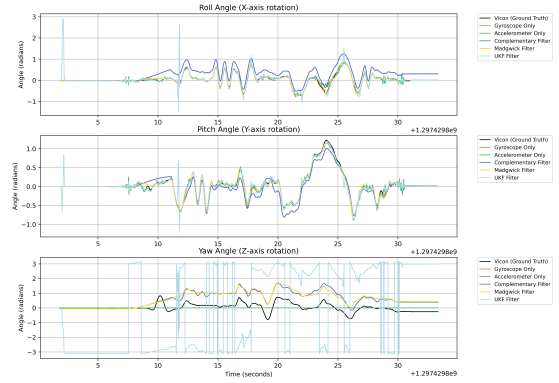


Fig. 6: Orientation comparison for Dataset 4: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

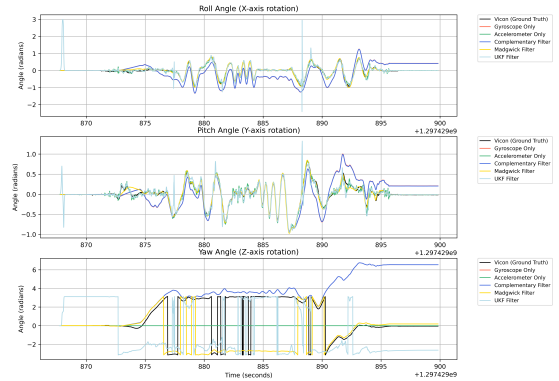


Fig. 7: Orientation comparison for Dataset 5: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

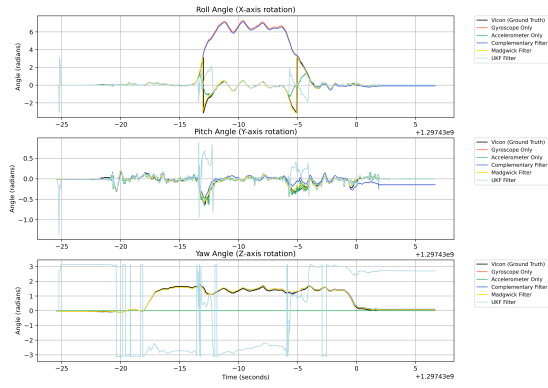


Fig. 8: Orientation comparison for Dataset 6: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

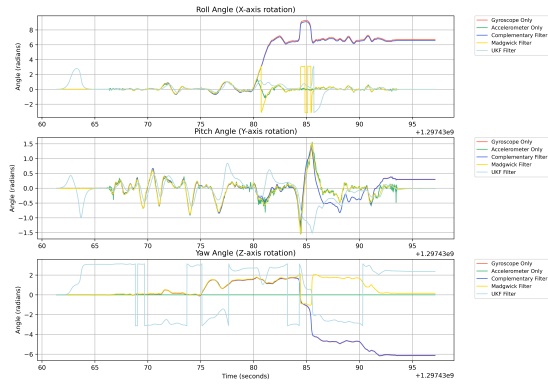


Fig. 9: Orientation comparison for Dataset 7: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

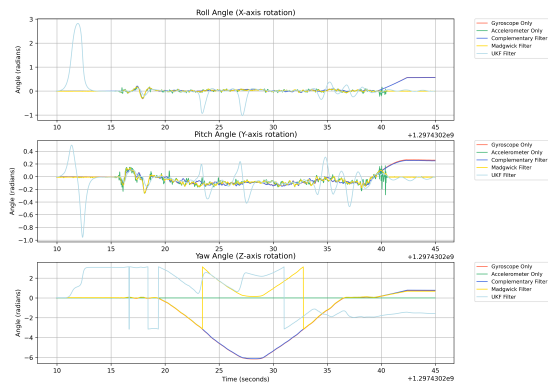


Fig. 10: Orientation comparison for Dataset 8: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

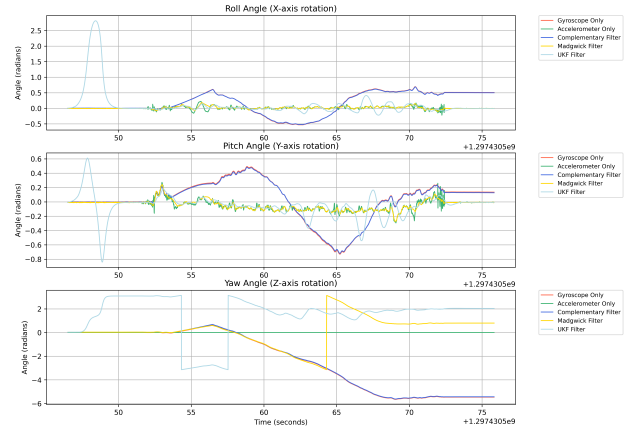


Fig. 11: Orientation comparison for Dataset 9: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

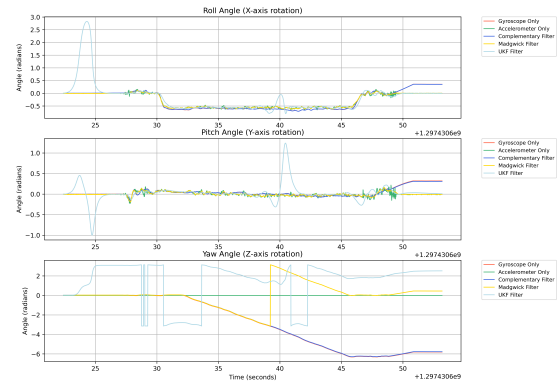


Fig. 12: Orientation comparison for Dataset 10: Orientation from Vicon, gyro, accelerometer, complementary filter, madgwick filter and UKF

## V. CONCLUSION

This work demonstrates the implementation and evaluation of three complementary methods for 3-D orientation estimation using a 6-DoF IMU. Gyroscope integration provides smooth short-term motion tracking but suffers from drift over time, while accelerometer-based inclination is reliable under static conditions but noisy during dynamic motion. The complementary filter effectively fuses these two sources, leveraging the accelerometer for low-frequency stability and the gyroscope for high-frequency responsiveness, resulting in a robust and accurate orientation estimate. However, the use of Euler angles can suffer from gimbal lock, which can be overcome by representing orientation using quaternions providing a singularity-free 3-D rotation representation suitable for continuous tracking in robotics and motion capture applications. Extending this approach, the Madgwick filter employs a quaternion-based sensor fusion algorithm that efficiently combines gyroscope and accelerometer data. By integrating gyroscopic angular velocity for smooth orientation propagation and applying a gradient descent-based correction derived from accelerometer measurements, it effectively mitigates gyroscopic drift. This results in improved accuracy and responsiveness, particularly during dynamic motion, outperforming traditional complementary filtering methods. Furthermore, an Unscented Kalman Filter (UKF) was implemented to provide a probabilistic framework for orientation estimation. By propagating a set of sigma points through the nonlinear system dynamics and incorporating both gyroscope and accelerometer measurements, the UKF captures the underlying system uncertainty more accurately than linear approximations. This leads to improved robustness in the presence of sensor noise and dynamic maneuvers, making it well-suited for real-world robotics and navigation applications.

## ACKNOWLEDGMENT

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